Explosive Cyber Security Threats During COVID-19 Pandemic and a Novel Tree-Based Broad Learning System to Overcome

Ying Gao, Hongyue Miao, Jixiang Chen, Binjie Song, Xiping Hu, and Wei Wang

***Abstract*— The rapid spread of the COVID-19 has not only affected personal health and economy, but also revolutionized people's lifestyles. As more people turn to work and socialize online, the development of unmanned technologies based on the Internet of Vehicles (IoV), such as unmanned delivery, unmanned vehicles, unmanned transportation, etc., will become an inevitable trend. However, all kinds of intelligent terminals for unmanned equipment require a large amount of data interaction with devices such as cloud servers, mobile terminals, and roadside terminals, which poses cyber security risks. Furthermore, the outbreak of COVID-19 has prompted people to put forward higher demands for the security of network communications. Unfortunately, the current intrusion detection methods based on machine learning still have weaknesses such as low accuracy and low efficiency when faced with unbalanced data distribution. To solve the above problems, we propose a novel Tree-based BLS (TBLS) intrusion detection method according to the idea of ensemble learning and decision tree (CART and J48). The performance of TBLS was tested on the NSL-KDD dataset and the UNSW-NB15 dataset respectively, which contain a variety of malicious traffic types for attacks on the IoV. The results show that our proposed method can achieve higher accuracy rate and lower false alarm rate, compared with the existing 16 solutions.**

***Index Terms*— Intrusion detection, broad learning system (BLS), decision trees, ensemble learning.**

I. Introduction

SINCE 2020, the number of COVID-19 cases has been increasing and spreading rapidly to the world. It has not only affected personal health and economy, but also completely changed people's ways of life. More people choose to work and socialize online to protect themselves from the virus. Thanks to huge advances in artificial intelligence, mobile sensing, ultra-reliable and low-latency communications, and intelligent manufacturing technology, the Internet of Vehicles (IoV), which refers to a vast interactive network composed of information such as vehicle position, speed, and routes, has grown rapidly in recent years, especially during the COVID-19 pandemic. The great breakthrough in unmanned technologies has not only attracted the attention of industry and academia, but also brought potential network security risks, because all intelligent terminals of unmanned equipment need to interact with cloud servers, mobile terminals, roadside terminals, and other devices for large amounts of data interaction [1]–[4].

Taking the VANET as an example, a distributed denial of service (DDoS) attack may occur during vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication [5]. Once the intrusion is successful, it may cause communication congestion, or even send malicious instructions to mislead the vehicle's actions, and even lead to serious traffic accidents. Monitored by the Ministry of Industry and Information Technology (MIIT) of China, there were more than one million malicious attacks appeared on the VANET platforms in the first half of 2021, which is an increase of more than 80% over the same period in 2020. With the spread of COVID-19, people have put forward higher requirements for the security and stability of the IoV [6].

Nowadays, experts have done lots of research work [7]–[10] to improve the safety and reliability of IoV, adopting mainly machine learning-based methods, such as artificial neural networks (ANN) [11], convolutional neural networks (CNN) [12], decision trees (DT) [13], random forests (RF) [14], support vector machines (SVM), Naive Bayes (NB) [15] and so on. However, the current intrusion detection methods expose the following challenges [16]:

1) Anomalous traffic is relatively difficult to obtain, which results in more normal data in the training set and fewer anomalous data. The imbalance of the dataset leads to insufficient training of the detection method, and the detection results often tend to classify the abnormal traffic as normal.

2) There are differences between the same types of traffic, and similarities between the different types of traffic, which makes it difficult to detect anomalies.

3) It is not known in advance which features contribute greatly to anomaly detection.

4) VANET has high real-time performance and a large amount of data, which requires high efficiency in detection.

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At present, the anomaly detection method based on deep learning needs to be layer-by-layer in training. This results in large time overhead and low efficiency.

To improve the accuracy of intrusion detection, the existing methods usually take two approaches [7]. One is to obtain better feature representation in order to enlarge the difference between data distributions and the other is to enhance the recognition ability of the classifier.

Although these approaches have achieved good intrusion detection results, in fact, most of them are still misuse-based method rather than advanced. With the continuous increase of data volume, the demands for data processing efficiency will also become higher. Thus, complex methods such as ensemble learning [17]–[19] were adopted to improve the performance of the classifier. However, the added time and space overhead of these methods will limit their deployment on NIDS.

Considering the existing problems comprehensively, this paper firstly proposes an intrusion detection method based on BLS. As far as we know, this is the first time that BLS [20] is applied to intrusion detection. We also propose a novel method named Tree-based BLS (TBLS) to detect network intrusion in IoV systems. The implementation process is as follows. First, multiple decision tree classifiers are used to fit the features of the dataset. Second, based on the idea of ensemble learning, we use BLS to integrate the features fitted by each basic tree classifier, and finally get the distribution features of the raw traffic. In the intrusion detection stage, BLS performs multiple non-linear transformations on the input nodes to obtain feature nodes. Then the input nodes and feature nodes are concatenated for the next step and the final classification result is obtained through the pseudo-inverse operation.

The contributions of this paper include the following aspects:

1) To our best knowledge, this is the first time that BLS has been adopted in network intrusion detection. This paper realizes the preliminary exploration of BLS in intrusion detection and shows how to set various parameters in order to maximize its effectiveness. We also have provided references for the latter research work.

2) We have improved the BLS intrusion detection method and proposed the TBLS method, which uses trees to improve the input nodes of the original BLS and further improve the generalization ability of the BLS.

3) The experimental results show that the proposed method can adapt to intrusion detection better. Compared with other proposed intrusion detection method, TBLS can obtain higher accuracy and a lower false alarm rate, and it shows higher intrusion detection efficiency in the test phase.

The structure of the following paper is as follows: Section II mainly introduces the related work of the paper. Section III is the key part, which mainly describes the BLS and TBLS algorithms in detail. Section IV includes the analysis of the experimental content and results. The summary and future work are in Section V.

II. Related Work

As we know, network intrusion detection systems (NIDS) can be divided into two categories [21]. The first category is the misuse-based anomaly detection method. This method creates a library of exception rules, which stores known types of attack behaviors. During detection, if the network traffic matches a rule in library B, the traffic is identified as abnormal. Thus, this method cannot detect unknown abnormal traffic. The other category is the anomaly-based detection method, which only learns the pattern of normal network traffic. During detection, if the network traffic cannot be identified by the model, the traffic will be regarded as abnormal. Although this method can identify new types of abnormal traffic, the false alarm rate is high because it will treat new normal traffic and traffic never seen before as an exception.

With the development of big data technology [22]–[25] and artificial intelligence (AI) [26]–[28], the self-learning model based on massive data is widely used in the field of NIDS. Nowadays, machine learning has been widely used in intrusion detection and can be divided into three categories.

The first category is supervised anomaly detection method. In the training process, a training set and labels representing categories are required. Shone *et al.* [29] combined the deep and shallow learning for intrusion detection. This approach adopted a deep neural network for feature extraction with random forest for classification. The result showed that it was more efficient and effective than the deep brief network (DBN). Yin *et al.* [30] used recurrent neural networks to build IDS that suitable for binary and multiclass classification. The experimental results showed that its performance is superior to that of traditional ML classification methods.

The second is unsupervised anomaly detection method. It requires only a training set and no labels. The classification is achieved by means such as clustering. Cao *et al.* [31] proposed a novel method to help the AE achieve shallow representation of abnormal traffic, and proposed two models which are Shrink Auto-encoder (SAE), and Dirac Delta Variational Auto-encoder (DVAE). They adopted seven one-classifiers to detect abnormal traffic. Experiments showed that SAE and DVAE can achieve higher accuracy when compared with the other three shallow representation methods. Ahmed and Mahmood [32] presented a clustering-based collective anomaly detection approach to detect denial of service (DoS) attacks in the traffic. Their approach was proved to be more efficient than other traditional cluster methods.

The third is the semi-supervised anomaly detection method. This method combines the above two methods and adopts a few labels to train the model. Semi-supervised learning methods can be further subdivided into transductive learning and inductive learning. An online sequential and network Entropy estimation based semi-supervised ML approach was proposed for DDoS detection in [33]. The supervised part was designed to reduce the false alarm rate of the unsupervised part. A semi-supervised learning method called FSSL-EL was also proposed to detect abnormal network traffic in [34]. After training the supervised part, the author adopted the fuzziness-based method to mine the hidden structure of unlabeled data and combined them via a neural network. However, the prediction performance of the above methods still needs to be improved.

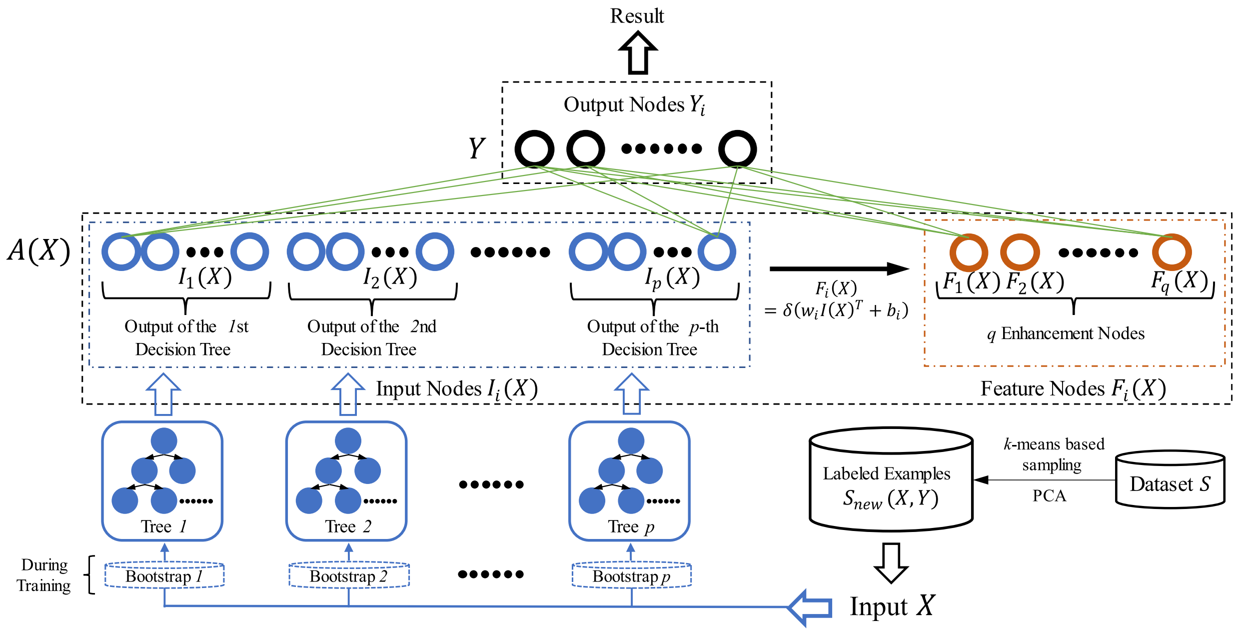

Fig. 1. Structure of TBLS.

III. The Implementation Process

A novel Tree-based Broad Learning System (TBLS) for intrusion detection is presented in this section, which can give a more accurate classification with less time consumption. The main structure of broad learning is also artificial neural network. However, unlike the deep learning system which mainly learns hidden features by increasing the number of layers in its neural network, a broad learning system adds neurons to each layer to extend features in order to realize thorough analysis and recognition of complex and multivariate data. Moreover, in order to further improve the fitting degree of traffic data and the processing performance of unbalanced data, we have involved several weighted DT classifiers in our structure.

In the following subsections, data preprocessing, k-means based data sampling, broad learning system and the novel tree-based BLS method for NIDS will be introduced respectively in detail. For better understanding our method, the flowchart of the proposed approach is described in Fig.1.

A. Data Preprocessing

In this paper, we set $X = \{x_1, x_2, \ldots, x_n\} \in R^{m\times n}$ to denote the training traffic with features, where n expresses the amount of traffic in the training dataset, and m represents the number of dimensional features. x_i^j represents the j-th feature in x_i.

Due to the existence of non-numerical features in X which may cause a negative effect on the learning process, we adopt the one-hot encoding method to transform symbolic features into numerical data. For example, if a column in X has three distinct values $\{f_1, f_2, f_3\}$, and then they will be mapped to three binary vectors $[1, 0, 0]$, $[0, 1, 0]$, $[0, 0, 1]$ respectively, which are regarded as their new values.

After encoding, we use min-max standardization to scale all the feature values within the interval [0, 1], the new value is calculated by:

$$\widehat{x_i^j} = \frac{x_i^j - MIN^j}{MAX^j - MIN^j} \tag{1}$$

where $\widehat{x_i^j}$ is the attribute value, MIN is the minimum of the j-th column while MAX is the maximum.

Meanwhile, $Y = \{y_1, y_2, \ldots, y_n\}$ denotes the label of X, and the range of y_j is determined according to the classification categories. For instance, we set $y_i \in \{0, 1\}$ if there are only two types of records: normal and abnormal. If there are five types to be identified, we will set $y_i \in \{0, 1, 2, 3, 4\}$. We also implement the one-hot encoding method on Y as well, it will facilitate the training process of the whole model.

In addition, the redundancy and correlation between features in Traffic data may slow down the training process and reduce detection efficiency. Hence, we use the Principal Component Analysis (PCA) algorithm to reduce the dimensions of the dataset for better feature selection and improving the speed of training process. Principal Component Analysis (PCA) is a kind of unsupervised learning feature extraction method which maps high-dimensional data to a low-dimensional space by linear projection. Useful information will be kept after the change. In this paper, we adopt PCA to discard the redundant components and extract the important features.

B. K-Means Based Data Sampling

A universal characteristic of the network traffic dataset is that the original data is unbalanced. The number of normal data is always much larger than that of abnormal data. Thus, once a classifier is trained on an imbalanced dataset, it will tend to bias towards those frequent records. To avoid this,

Algorithm 1 K-Means Based Sampling Approach

Input: The labeled examples $S = \{X, Y\}$, the number of categories C, the number of clusters k, and sampling threshold n_{over}
Output: The labeled examples S_{new} after sampling

initialize $S_{new} = \{\}$
for $i = 1 : C$
  select all examples that labeled i from S, and mark them as $S^i = \left\{X^i, Y^i\right\}$
  if $\left|S^i\right| \leq n_{over}$
    randomly select $n_{over} - \left|S^i\right|$ records from S^i as S^{dup}, add perturbation to X^{dup} in S^{dup}
    $S_{new} = S_{new} \cup S^{dup} \cup S^i$
  end if
  if $\left|S^i\right| > n_{over}$
    calculate sample rate $r = n_{over} / \left|S^i\right|$
    fit a k-means model with k and X^i
    get the clustered examples set $\left\{S_1^i, \ldots, S_k^i\right\}$
    for $j = 1 : k$
      randomly select $\left\lceil \left|S_j^i\right| * r \right\rceil$ records to form $\hat{S_j^i}$
      $S_{new} = S_{new} \cup \hat{S_j^i}$
    end for
  end if
end for
return S_{new}

we adopt a k-means based sampling method to get a balanced training set from an imbalanced one.

First, we define n_{over} as the sampling threshold, k as the number of clusters for a k-means clustering algorithm, S_{new} as the sampled dataset. Second, we divide the training set S into C subsets $S^i (i \in \{1, 2, \ldots, C\})$ based on the label. For a subset S^i, if the number of records is less than n_{over}, we randomly select $n_{over} - \left|S^i\right|$ records from S^i, then add noise to those duplicates and put both the duplicates and the original subset into S_{new}. For a subset whose quantity is more than n_{over}, we first cluster the subset into k clusters and calculate the sample rate r by a function which is defined in Algorithm 1, then randomly select records from each cluster at this sample rate and add them into S_{new}. Finally, we get a new dataset S_{new}, which is a balanced one with around $C * n_{over}$ records. The details of the sampling approach are presented in Algorithm 1.

After data sampling, if n_{over} is set appropriately, the new training set can well represent the raw dataset and is suitable for training a fairer classifier.

C. Broad Learning System (BLS)

Deep learning-based method includes many hidden layers. It can further grasp the correlation between features and the target labels by continuously increasing the depth of the structure. However, the huge number of parameters to be optimized in a deep learning structure usually takes a lot of time and machine resources. To resolve these issues, the Broad Learning System was first introduced in [20], aiming to provide an alternative approach to deep learning. In order to save the time consumption caused by multi-layer neural network, there are only two layers in a BLS. The first layer contains input nodes together with feature nodes which are mapped by input nodes. And the second layer is the output layer, which has the same number of nodes as the number of categories. A BLS can expand the model by adding feature nodes to the first layer, and in this way extract more useful and non-linear information. Besides, the BLS does not require backpropagation to find the optimal weights because it only has two layers. Instead, Pseudoinverse and ridge regression learning algorithms are adopted to calculate the optimal weights.

Assume the input data is X, the feature nodes group is $F(X) = \left[F_1(X), F_2(X), \ldots, F_q(X)\right]$, the number of the feature nodes is q, the i-th feature node is calculated via the following formula:

$$F_i(X) = \delta(w_i X + b_i) \tag{2}$$

where $w_i (i = 1, 2, \ldots, q)$ indicates weight and $b_i (i = 1, 2, \ldots, q)$ indicates bias, which are both randomly generated. $\delta(\cdot)$ represents the activation function. In this paper, we adopt $tanh$ function as the activation function. The formula is as follows:

$$\delta(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{3}$$

After calculating the feature nodes from the input, the first layer $A(X)$ of the BLS can be expressed as:

$$\begin{aligned} A(X) &= [X|F(X)] = [X, F_1(X)] \\ &= \left[X, F_1(X), \cdots, F_q(X)\right] \end{aligned} \tag{4}$$

Finally, calculate the weight matrix W which satisfies the equation $A(X) W = Y$. Because there are only two layers in BLS, the optimal weight can be calculated by a simple inversion method. The formula is as follows:

$$W = A(X)^{-1} Y \tag{5}$$

However, the inverse matrix of $A(x)$ may not exist, and we cannot get $(A(X))^{-1}$ directly. Therefore, a pseudo-inverse operation is adopted to substitute the matrix inverse operation, expressed as $A(X)W \approx Y$. The pseudo-inverse of a matrix M can be calculated by the following formula:

$$M^+ = \lim_{\lambda \to 0} (M^T M + \lambda I)^{-1} M^T \tag{6}$$

the pseudo-inverse of $A(X)$ and the optimal weight matrix W are calculated as follows:

$$A(X)^+ = \lim_{\lambda \to 0} (A(X)^T A(X) + \lambda I)^{-1} A(X)^T \tag{7}$$

$$W = A(X)^+ Y \tag{8}$$

It is obviously that it takes much less time to get the weight matrix compared with deep learning approaches which require complex backpropagation processes. Thus, the BLS shows better real-time performance in dealing with the massive dataset of VANET and at the same time effectively avoids the transmission error between layers in backpropagation.

D. Tree-Based BLS Method for NIDS

Since all the feature nodes are generated from the input X, the quality of the input data plays a vital role in the performance of BLS. While the raw data of traffic is not able to fully present the relationship between feature X and labels Y, we reconstruct the input nodes of BLS with the help of ensemble learning, and put forward a novel network intrusion detection method named Tree-based BLS (TBLS). The TBLS uses the extended feature nodes of broad learning algorithm to integrate the learning results of each weighted decision tree classifier and adaptively calculate the input weight of each classifier so that it can achieve better predictions than a single classifier.

Assume the input of BLS is $I(X)$, for an original BLS, $I(X) = [X]$. Instead of using raw training data X as input, we define $I(X) = [I_1(X), I_2(X), \ldots, I_p(X)]$, where I_i is a decision tree classifier, p denotes the number of the trees, and $I_i(\cdot)$ is the i-th tree's decision function, $i \in \{1, 2, \ldots, p\}$. The dimension of $I(X)$ is $p \times C$, where C denotes the number of classes.

First, for the training dataset $S = \{X, Y\}$, we adopt a bootstrap sampling strategy to get some samples from S as B_i to train the i-th basic classifier I_i. The sample rate r is set to 0.4 based on sufficient experimental results in this paper. After p classifiers have been generated, input X to each basic classifier to get the predictions, and assemble them to form the input $I(X) = \left[I_1(X), I_2(X), \ldots, I_p(X)\right]$ of BLS.

Second, according to the training approach of BLS described in the previous section, the i-th feature node is mapped via the following formula:

$$F_i(X) = \delta\left(w_i I(X)^T + b_i\right) \tag{9}$$

thus, the first layer can be present as:

$$A(X) = [I(X)|\mathrm{F(X)}] = \left[I(X), F_1(X), \cdots, F_q(X)\right] \tag{10}$$

where q denotes the number of the feature nodes.

Finally, the weight matrix W which satisfies the equation $A(X)W = Y$ is calculated by the following formula:

$$W = A(X)^{+}Y \tag{11}$$

where $A(X)^{+}$ is the pseudo-inverse matrix of $A(X)$.

In the testing phase, for a record x, after data preprocessing, the input is mapped to $I(x) = [I_1(x), I_2(x), \ldots, I_p(x)]$ in the first step. Then, use all the w_i and b_i which have been fitted during the training process to build the i-th feature node and $F(x)$. After that, $I(x)$ and $F(x)$ are used for generating $A(X)$ via Eq. (10). Finally, the output of x is $\hat{y} = A(X)W$. Since there are C nodes in the output layer, $\hat{y}$ contains C values, and the label corresponding to the maximum value of $\hat{y}$ is considered as the prediction class.

To make it easier to understand, the details of the TBLS algorithm are demonstrated in Algorithm 2.

Algorithm 2 Tree-Based BLS

Input: The labeled training dataset $S = \{X, Y\}$, the number of the basic classifiers p, the number of the feature nodes q and the bootstrap sample rate r
Output: Tree-based BLS model χ
1: $\chi = \{\}$
2: **for** $i = 1 : p$
3: Generate bootstrap B_i by sampling S with sample rate r
4: Train a tree classifier I_i on B_i, $\chi = \chi \cup \{I_i\}$
5: **end for**
6: $I(X) = \left[I_1(X), \ldots, I_p(X)\right]$
7: **for** $i = 1 : q$
8: randomly initialize weight w_i and bias b_i
9: $\chi = \chi \cup \{w_i, b_i\}$
10: Calculate $F_i(X)$ use Eq. (9)
11: **end for**
12: $F(X) = \left[F_1(\mathrm{X}), \ldots, F_q(\mathrm{X})\right]$
13: $A(X) = [I(X)|F(X)]$
14: Calculate the optimal weight W use Eq. (11)
15: $\chi = \chi \cup W$
16: **return** χ

IV. Experiment and Analysis

A. Evaluation Measures

In our work, accuracy (AC), detection rate (DR), and false alarm rate (FAR) are used to evaluate the performance of the proposed model. The representations of these three metrics are given as follows:

$$AC = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

$$DR = \frac{TP}{TP + FN} \tag{13}$$

$$FAR = \frac{FP}{FP + TN} \tag{14}$$

where true positive (TP) refers to the number of attack records correctly identified as an attack, true negative (TN) denotes the number of normal records that are correctly identified as normal, false positive (FP) denotes the number of normal records incorrectly identified as an attack, and false negative (FN) refers to the number of attack records incorrectly identified as normal. The definition of a confusion matrix is shown in Table I. The accuracy describes the ability of a detection model to make correct prediction while the detection rate indicates the probability that attacks can be identified correctly. The false alarm rate indicates the rate that the normal records are predicted as attacks. Researchers try their best to build a detection model with high accuracy and low FAR.

TABLE I
CONFUSION MATRIX

		Predicted Class	
		Normal	Attack
Actual Class	Normal	TN	FP
	Attack	FN	TP

B. Experiment Dataset and Settings

KDD Cup 99 dataset has been widely used as a benchmark dataset for evaluating the performance of IDS for many years.

TABLE II
THE DISTRIBUTION OF CLASSES IN NSL-KDD DATASET

	Training Dataset		Testing Dataset	
	KDDTrain	**KDDTrain_20percent**	**KDDTest+**	**KDDTest-21**
Normal	67343	13449	9711	2152
Probe	11656	2289	2421	2402
DoS	45927	9234	7458	4342
U2R	52	11	200	200
R2L	995	209	2754	2754
Total	125973	25192	22544	11850

TABLE III
THE DISTRIBUTION OF CLASSES IN UNSW-NB15 DATASET

	Training-set	**Testing-set**		**Training-set**	**Testing-set**
Normal	37000	56000	**Fuzzers**	6062	18184
Reconnaissance	3496	10491	**Worms**	44	130
Backdoor	583	1746	**Shellcode**	378	1133
DoS	4089	12264	**Generic**	18871	40000
Exploits	11132	33393	**Total**	82333	175342
Analysis	677	2000			

However, it was observed that the training data contains a lot of redundant records. This redundancy causes the learning algorithms to be biased towards the frequent records and leads to poor classification results of harmful records. Thus, this paper is tested on the other two common datasets, NSL-KDD [35] and UNSW-NB15 [40], respectively.

NSL-KDD dataset is an improved version of the KDD Cup 99 dataset, and it was built by eliminating the redundant records from the training and test data in KDD Cup 99 dataset. Hence, the NSL-KDD dataset contains a reasonable number of records, which make it more suitable for the training and evaluating of any learning algorithm. Nowadays, although there are some other newer datasets available for evaluation, NSL-KDD dataset is still considered as one of the most authoritative benchmark datasets in this filed. Therefore, we utilize the NSL-KDD dataset in our work as mentioned before. The NSL-KDD dataset covers two training sets 'KDDTrain' and 'KDDTrain_20percent', 'KDDTest+' and 'KDDTest-21' as the testing sets. Each record consists of 41 features in numerical or symbolic type, and one class label for indicating whether the record is normal or a particular kind of attack. There are five typical network events involved in the dataset, i.e., normal, probe, Denial-of-Service (DoS), user to root (U2R) and remote to local (R2L). We provide Table II below to show the distribution of classes in NSL-KDD dataset.

The 'KDDTrain_20percent' dataset contains 20% records in the 'KDDTrain', which can still represent the distribution well. Specifically, we use the 'KDDTrain_20percent' dataset as our training dataset for improving training efficiency while guaranteeing a good performance. Besides, 'KDDTest+' and 'KDDTest-21' are used for testing.

The UNSW-NB15 dataset is collected and established by the Australian Centre for Cyber Security (ACCS) from IXIA PerfectStorm. The dataset contains multifarious normal and abnormal behaviors of traffic data. Compared with NSL-KDD, the UNSW-NB15 dataset has more novel types of attacks and features. It has nine aggressive behaviors in total: Fuzzers, Analysis, Backdoor, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms. The raw data of UNSW-NB15 includes 47 features, and can be divided into two subsets 'UNSW-NB15_training-set.csv' and 'UNSW-NB15_testing-set.csv'. And in these subsets, the number of features has been reduced to 42. Table III shows more information about the distribution of classes in UNSW-NB15 dataset.

TABLE IV
THE PARAMETERS IN DATA PREPROCESSING

	NSL-KDD
PCA Dimensions d	10, **20**, 30, 40, 50, 60, No Reduction
n_{over}	100, **400**, 800, 1200, 1600
	UNSW-NB15
PCA Dimensions d	10, 20, **30**, No Reduction
n_{over}	1000, 2000, **3000**

The experiments are performed on a personal computer, and the configurations are listed as follows: The Windows 10 professional system, Intel Core i7-8700 CPU @ 3.20 GHz, and 16 GB RAM, NVIDIA GeForce RTX 2070S GPU.

C. Experimental Results

In order to make our experimental testing process easier to understand, all the specific parameters of the experiments under the NSL-KDD dataset and UNSW-NB15 dataset are written in the following Table IV and Table V.

1) The Impact of the PCA Method: To maximize the performance of the proposed model, we designed an experiment

TABLE V
THE PARAMETERS OF TBLS

	Tree-based BLS
Sample rate r	0.10, 0.25, **0.40**, 0.55, 0.70, 0.85
Number of trees p	20, 40, 60, 80, **100**
Number of feature nodes q	20, 40, 60, **80**
Basic classifier (weighted)	**J48**, CART, C4.5, GNB

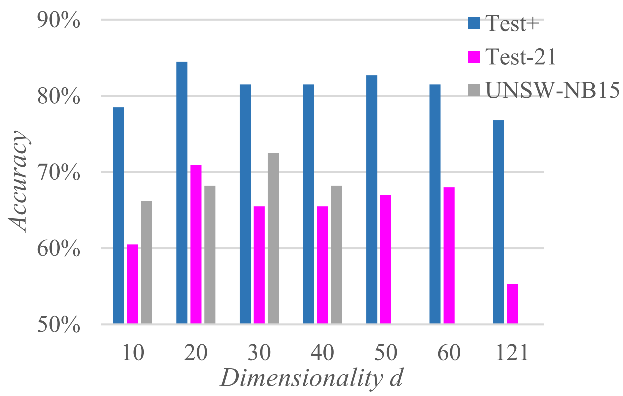

Fig. 2. Relationship between PCA dimensionality d and detection accuracy.

to find out the optimal parameter for PCA. The results of NSL-KDD and UNSW-NB15 are shown in Fig.2. For the NSL-KDD dataset, when the dimensionality is set to 20, the proposed model reaches its best performance in terms of accuracy on Test+ and Test-21 datasets. When the dimension is lower than 20, the processed data is not able to describe the raw data fully, and the performance is unsatisfactory predictably. Meanwhile, when the dimension is higher than 20, the processed data may generate irrelevant data that adversely affect the detection performance and consume more computing resources. When the dimension is set to 121, it means the PCA is not applied, and we have observed that the accuracy drops sharply. For the UNSW-NB15 dataset, the model peeks when the dimensionality is set to 30. Neither higher than 30 nor lower than 30 can give better performance. The result shows that it is necessary to apply PCA to reduce the dimension of the data before the training process. Thus, we choose to use the 20-dimension and the 30-dimension data respectively for our proposed model in the following experiments.

Besides, we also made tests about the threshold n_{over} of the k-means based sampling method to explore the relationship between n_{over} and detection accuracy. Fig.3 shows the whole results of the above two datasets. When n_{over} is too small, the lack of data samples results in underfitting, which makes the model unable to achieve a satisfactory training effect. Meanwhile, when n_{over} is too large, the algorithm is more inclined to classify traffic into normal class and ignore other aggressive behaviors. Therefore, it is proved that proper sampling can reduce the impact of data imbalance to a certain extent, make the algorithm pay more attention to those attack classes, and then improve the performance of data prediction and detection of attack behaviors. As a result, we decided to choose 400 and

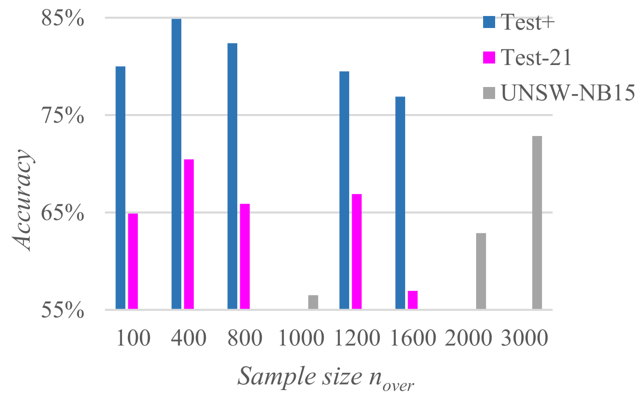

Fig. 3. Relationship between sample size n_{over} and detection accuracy.

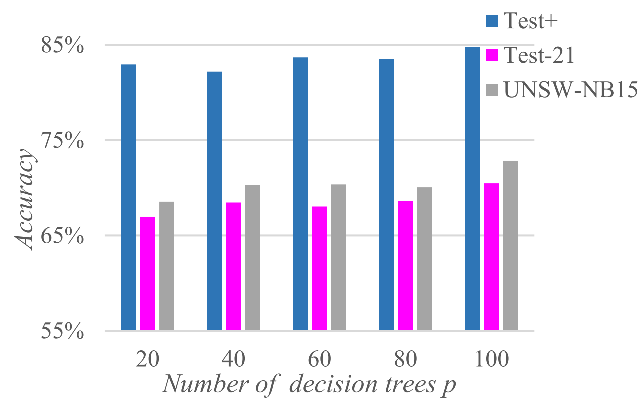

Fig. 4. Relationship between number of decision trees p and detection accuracy.

3000 as the sample size of the NSL-KDD dataset and the UNSW-NB15 dataset respectively.

2) The Impact of the Bootstrapping Method: In this part, we test the effects of two parameters in the bootstrapping process which are the number of decision trees p and the sample rate r, to study the efficiency of the ensemble learning method. Relative results are shown in Fig.4. When p is set to 100, the TBLS method performs the best. As p declines from 100, there is a significant decrease in accuracy, which indicates that the model is underfitting and cannot learn the training data comprehensively. Therefore, we choose $p = 100$ as the optimal number of decision trees.

Furthermore, it is shown in Fig.5 about the performance of the TBLS algorithm when different sample rates are used. Through the outcomes, when r is set to 0.4, the model gets the best accuracy in all three datasets. If the r is set below 0.4, the accuracy of the model will be poor due to lack of sampled data. However, the performance of the model will still decline when r is set over 0.4, since TBLS cannot fully learn the distribution of the dataset based on a mass of data. As a result, we choose $r = 0.4$ as the sample rate of our approach.

3) The Impact of the Feature Nodes: In order to verify the effect of increasing feature nodes in broad learning system on the overall performance of model, we change the number of feature nodes q to see which is the most appropriate one. The

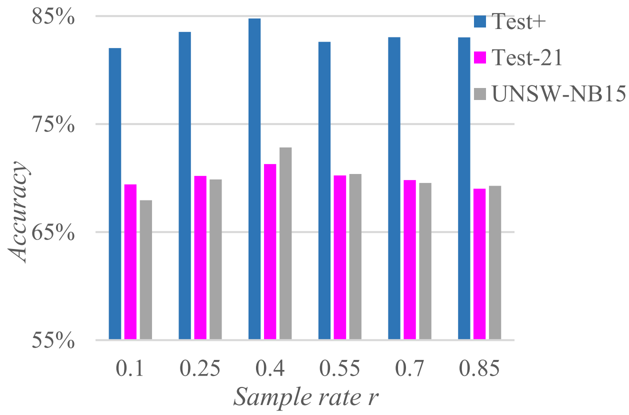

Fig. 5. Relationship between sample rate r and detection accuracy.

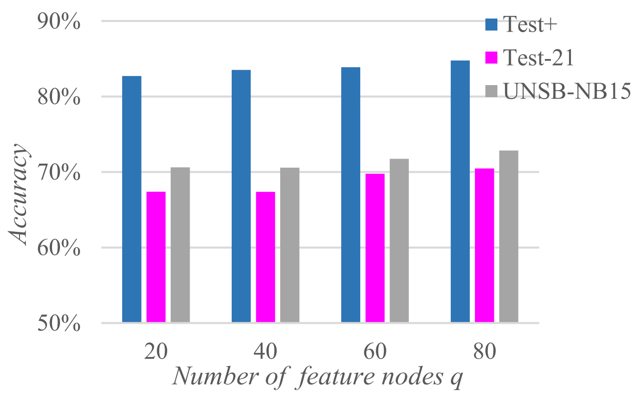

Fig. 6. Relationship between number of feature nodes q and detection accuracy.

results are shown in Fig.6, illustrating that the detection ability to traffic data is constantly improving with the increase of the feature nodes. However, limited by the personal computer's performance in the experiment, we can only set q to 80 as the optimal number of feature nodes.

4) The Impact of the Basic Classifier: To explore the influence of different basic classifiers on the detection accuracy of our TBLS model, we tried three decision tree classifiers J48, CART and C4.5 respectively as the basic classifiers. Besides, an original BLS based on GNB classifiers was built to act as the baseline for comparison. For each model, grid search and cross-validation techniques are used to search for the best parameters. After training, we apply the models to the testing set for multiple classifications.

The results are described in Fig.7. The detecting accuracy of BLS(GNB) in 'KDDTest-21' and 'UNSW-NB15_testing-set' drops to almost 50% without the help of the decision trees, and the integrated model gets the highest accuracy score when J48 serves as the base classifier. In a word, the TBLS has proved to be more efficient than the original BLS for intrusion detection with a stronger non-linear representation produced by the basic decision tree classifiers. Moreover, after weighted processing, the decision tree classifiers can improve the performance of the whole model more effectively.

5) Comparison With Other Existing Methods: In this section, we compare the proposed method with 16 existing methods. Among them, 'Experiment-1' in [38] and 'Experiment-2' in [39] are methods based on semi-supervised learning. In addition, we also used the original BLS framework and the tree-based deep learning system (TDLS) to conduct ablation experiments. The results are shown in Table VI.

The accuracy of TBLS is 84.49% on 'KDDTest+', 70.92% on 'KDDTest-21', and 72.84% on 'UNSW-NB15_testing-set' (which is not listed). When compared with the most popular semi-supervised learning algorithms, our approach can still stay ahead. Although the accuracy of TBLS is relatively weak on 'KDDTest+', it has obvious advantage in FAR, which is regarded more important in the field of intrusion detection.

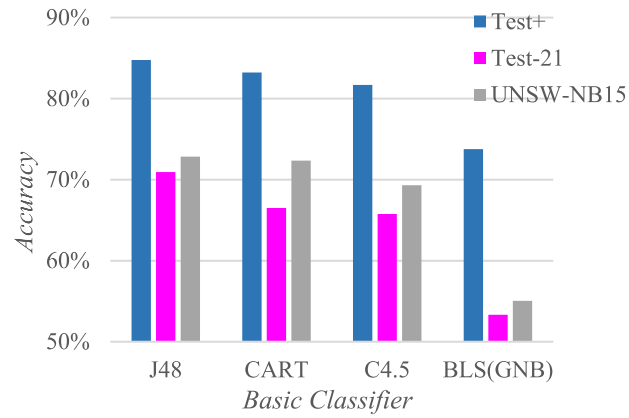

Fig. 7. Classification results of different basic classifier.

TABLE VI

COMPARED WITH THE CURRENT DETECTION METHODS

Methods	KDDTest+		KDDTest-21	
	AC	FAR	AC	FAR
J48[35]	81.05%	N/A	63.97%	N/A
NB[35]	76.56%	N/A	55.77%	N/A
NB Tree[35]	82.02%	N/A	66.16%	N/A
RF[35]	80.67%	N/A	63.25%	N/A
RT[35]	81.59%	N/A	58.51%	N/A
MLP[35]	77.41%	N/A	57.34%	N/A
SVM[35]	69.52%	N/A	42.29%	N/A
STL[36]	79.90%	20.10%	N/A	N/A
DNN[37]	75.75%	N/A	N/A	N/A
RNN-IDS[30]	81.29%	N/A	64.67%	N/A
CNN[12]	79.48%	N/A	60.71%	N/A
AE+GNB[30]	83.34%	N/A	N/A	N/A
Experiment-1[38]	82.41%	N/A	67.06%	N/A
Experiment-2[39]	84.12%	N/A	68.82%*	N/A
DBN[39]	80.58%	19.42%	N/A	N/A
S-NDAE[29]	**85.42%****	14.58%*	N/A	N/A
BLS	68.96%	11.13%	42.05%	47.38%
TBLS (Ours)	**84.49%**	**6.14%**	**70.92%**	**23.62%**
TDLS	82.96%	6.92%	69.74%	25.32%

The great improvement of our TBLS over the original BLS have emphasized the importance of using tree structure as the basic classifier. Furthermore, in order to verify the superiority of BLS in NIDS, we replace the broad learning structure in Fig.1 with a 3-layer neural network to connect each decision tree classifier. Momentum-based gradient descent algorithm was used to train the new model called TDLS. Experimental results show that the TBLS not only learns faster, but also has better detection performance than TDLS in NIDS scenarios.

In sum, the proposed TBLS is superior to other methods regardless of AC or FAR. Results have already proved that our proposed method can be well adapted to intrusion detection.

V. Conclusion

In this paper, a novel Tree-based broad learning system (TBLS) method is proposed for intrusion detection task in VANET during COVID-19 pandemic. The presented TBLS approach achieves 84.49%, 70.92%, and 72.84% accuracy respectively on 'KDDTest+', 'KDDTest-21', and 'UNSW-NB15' datasets through a large series of experiments, which stand out from the other state-of-the-art methods.

Overall, the main contributions of our method can be summarized as follows. Firstly, it is the first time that BLS has been applied to intrusion detection domain as far as we know, aiming to reduce training time consumption and provide a faster way to remodel without retraining. Secondly, multiple decision trees are trained to reconstruct the input of a BLS so that the integrated TBLS model can perform better on multi-class classification. Finally, several approaches like PCA method and k-means based sampling method have been adopted to discard the redundant part of data for extracting important features and address the issue of data imbalance, which ensures the quality of data and thus raises the ceiling of performance.

In the future research, we are going to try more machine learning structures like the Generative Adversarial Networks in order to improve the generalization ability of our model and realize the real-time detection of new attacks. Our goal is to build an all-around intrusion detection system. By the way, we will test our approach on more public benchmarks.

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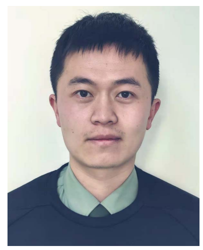

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